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Behaviours, Institutions, and the Architecture of Competitive-Social Networks

*A framework for describing, measuring, and comparing human
networks organized around shared practice*

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Abstract

This paper proposes a framework for describing human networks that form around a shared competitive or creative practice — esports scenes, sports fandoms, hobbyist communities, and adjacent structures. Its central claim is that such a network is better understood as a set of **behaviours people activate into and out of** than as a population divided into fixed segments ("players," "viewers," "fans"). A person who plays today may watch tomorrow and discuss both on the weekend; the same behaviours recur across what look like very different communities, which means they can serve as a common measurement language across networks that would otherwise seem incomparable. The framework defines a small set of **participation behaviours**, a companion concept of **institutions** (the persistent structures sustained behaviour crystallizes into), and a set of **dimensions** — institutionalization, parasocial attachment, social embeddedness — that cut across behaviours rather than belonging to any one of them. It argues that network "types" should be discovered by clustering real networks on these measurements, not assumed in advance, and that claims about which behaviour drives a network's growth should be tested empirically per network rather than built into the framework as an axiom. The paper closes with a discussion of how such a framework might be visualized, and situates the approach relative to existing literature in diffusion theory, parasocial relationship research, communities of practice, and institutional sociology.

The framework is proposed from applied work on a single network type — competitive video-game communities — measured through a single platform's creator-side trace data (23 titles, several weeks of continuous Twitch viewership data, supplemented by tournament-history records), and has not yet been validated against the other substrates its scope claims to cover. §9 closes with the specific empirical test that would close this gap, rather than leaving the limitation implicit.

1. Why this framework exists

Most existing ways of describing a competitive or creative community rely on a single proxy measurement, chosen because it happens to be the easiest thing to get historical data on, and then ask that proxy to answer a much bigger question than it was built for. A natural example: the age of an organized, prize-bearing competitive circuit around a given practice is often used as a stand-in for "when did this community mature." It's a reasonable instinct — professional competition is visible, dated, and institutionally documented in a way that grassroots participation usually isn't — but it quietly substitutes one narrow, late-arriving signal for a much broader question.

The substitution breaks in a specific, checkable way — but it is worth being precise about exactly what breaks, because the measure itself is not the problem. A professional competitive tier requires an existing base of participants large enough to produce elite performers and organizations willing to

invest in them; it cannot exist before that base does. So competitive-circuit age is necessarily a **lagging indicator of community formation** — it tells you when a community had *already* matured enough, for long enough, to support professionalization, not when the community itself formed or how large it actually was at the time. Call this claim (ii), and it is the one that fails.

Claim (i) is different, and does not fail: competitive-circuit age is a perfectly good *direct* measure of one specific thing — how institutionalized a community's competitive practice has become (§3.2 treats institutionalization as its own axis precisely because it is worth measuring on its own terms). The two claims get run together constantly because the same number answers both questions, well for one and badly for the other. The consequence of keeping them separate is not merely semantic: in applied work behind this framework, competitive-circuit age, read as claim (i), a direct institutionalization reading, outperformed a title's release year and franchise-continuity year as an explanatory variable for community structure — the opposite of what claim (ii)'s "merely a lagging proxy, therefore a weak measure" framing, taken alone, would predict. A lagging proxy for the wrong question can still be a strong direct measure of the right one.

A single scalar measure of either kind, in any case, can only describe one axis of what is, in practice, a multivariate situation: a community can be large and devoted to watching without ever developing an organized play tier, or it can have a thriving grassroots practice with no spectator culture to speak of. No single number — however well-measured, and whichever of the two questions above it actually answers — can summarize a state that genuinely has more than one dimension.

This paper's starting premise, then, is that the unit of analysis should not be "the competitive scene" treated as one thing, measured one way. It should be the **behaviours** that constitute a network's activity, measured separately and comparably, so that a network's actual shape — not a single borrowed proxy for it — becomes visible.

2. What kind of network does this describe

The framework describes what we call a **social-competitive network**: a group of people whose connection to each other and to a shared practice is constituted by what they *do*, not by a fixed role they occupy. Nobody is permanently "a player" or "a viewer" in this model — those are behaviours a person activates into for a bounded period, which is why the same person can occupy several of them in the same week, or the same evening.

This has a direct consequence for measurement. If behaviours are activated states rather than population segments, a network cannot be adequately described by headcounts of who belongs to which group. It has to be described by **readings on each behaviour, taken across the whole population, over time** — how much Play is happening, at what level of institutional organization, how much Watch, directed at what kind of source, and so on. The network's "shape" is the resulting profile across all of these readings, not a pie chart of member types.

A second consequence: because the same behaviour-vocabulary applies regardless of what the underlying practice actually is — a competitive video game, a traditional sport, a card game, a creative sandbox — networks that look superficially unrelated become genuinely comparable. The framework is deliberately agnostic about the delivery substrate. What matters is whether a persistent, competitive or creative, community practice exists around something, not whether that something runs on particular software.

3. The core architecture

3.1 Participation behaviours

A **participation behaviour** is a voluntary, individually performed act, bounded in time, whose performance is what constitutes a person's membership in the network at that moment. Behaviours are episodic and renewable, not permanent status changes — a person activates into Play today and may not tomorrow.

Working through a wide range of candidate behaviours and repeatedly applying a simple test — *is this a genuinely distinct act with its own output or goal, or is it better modelled as a property that modifies some other act* — narrows the list to five that survive as independent behaviours:

- **Play** — direct participation in the core practice itself, spanning casual through highest-stakes competing. Institutional context (see §3.2) is tracked separately rather than treated as a different verb.
- **Watch** — spectating, live or recorded.
- **Discuss** — commentary, analysis, and meta-discourse about the practice or its participants.
- **Attend** — synchronized physical co-presence at an event. (An earlier version of this framework also allowed "virtual attendance"; on inspection this collapses entirely into Watch at high social intensity and was removed — see §3.3.)
- **Create** — producing an artifact that becomes part of the network and would not otherwise exist: broadcasting oneself playing, writing a strategy guide, producing fan art, commentary and analysis content. These are differentiated by output type, not by separate verbs, because each is the same underlying act — composing a consumable output that others can then Watch, Discuss, or build on. (An earlier version of this list also included *organizing a tournament*. It does not survive the collapse test applied directly to it — see below.)

Several other candidates were considered and did not survive as independent behaviours, for three different reasons. "Elite play" and "professionalized play" are not separate acts from Play — they are the same act read at different points on the institutionalization dimension. "Streaming" is not a Play-variant but a Create-instance (it composes a broadcast output). "Organizing" an event — founding a league, running a tournament series — initially looked like a Create-instance too, on the reasoning that it produces an event-as-output. It fails the collapse test on closer inspection: Create's output is an

artifact that *accumulates toward* institutionalization over time (a body of strategy guides, a catalog of broadcasts), whereas organizing a tournament or league *is* the act of institution-formation itself, performed directly rather than approached asymptotically through artifact accumulation. Folding the two into one verb would erase the distinction §3.2 depends on, so organizing is treated as the originating act behind an institution rather than as a Create-instance — see §3.2. Two further candidates — Purchase and Socialise — looked at first like behaviours and turned out, on closer inspection, to be something else; both are discussed in §3.5.

What is currently measurable. The five behaviours above are symmetric in definition but not in observability, and that asymmetry deserves the same explicit treatment this paper gives the unmeasurable person-side dimensions in §3.3, rather than being left implicit.

Behaviour Observable from typical trace data?

Play	Yes — concurrent participation counts, match/session records
Watch	Yes — viewership figures, largely platform-native
Create	Partially — broadcast activity is directly observable; written guides, fan art, and most other artifact types generally are not, without a separate crawl of wherever they happen to be hosted
Discuss	Partially — chat and forum activity is observable where the platform exposes it, which is inconsistent and sometimes access-gated
Attend	Barely — ticket sales are rarely public; venue capacity is a ceiling on attendance, not a reading of it

This matters beyond honesty for its own sake. Any analysis built on this framework — including the precedence analysis in §5 — will be pulled toward whichever behaviours happen to have good data, which is exactly the failure mode §1 opens by criticizing, recurring one level up inside the framework meant to fix it. Naming the gap does not close it, but it keeps anyone reading results downstream from mistaking "well-measured" for "dominant."

A behaviour's readings co-occur rather than compete. A person attending a live event while competing is simultaneously doing Attend and Play; a person streaming themselves playing is simultaneously doing Play and Create. The framework does not ask "which behaviour is this person doing" as though the categories were mutually exclusive — it asks "what is this person's reading on each behaviour, right now," and several can be non-zero at once.

3.2 Institutions

An **institution** is a persistent, supra-individual, nameable structure — a league, a company, a team, a recognized tournament series, a community body — that forms when sustained participation behaviour crosses a threshold sufficient to justify an outside party (a publisher, a sponsor, an organizer, a platform) investing in formalizing it. Institutions outlive any single participant and any single instance of the behaviour that produced them. They have their own lifecycle — formation, growth, stabilization,

and eventual decline — and, critically, they feed back and reshape the future rate and shape of the very behaviour that produced them. A franchised, professionally organized competitive league, for instance, does not merely reflect an already-large player base; once it exists, it changes how players move between teams, how long careers last, and who enters the practice in the first place. Institutions are not passive residue. They are active participants in a network's subsequent evolution.

Institution is not a second category sitting beside behaviours in a parallel list. It is the **reified, named state a behaviour's own institutionalization reading produces once that reading crosses a threshold and holds**. Any measurement scheme that already tracks "when did organized competitive activity around this practice first reach a sustained, multi-region scale" is, in effect, already doing this — dating the formation of one specific institution — without necessarily recognizing it as one instance of a general mechanism that applies to every behaviour, not just Play.

This also settles where "organizing" belongs (§3.1). Founding a league or running a tournament series is not a Create-instance whose output later accumulates into an institution the way a growing body of strategy guides might. It is the act of institution-formation itself, performed directly — which is exactly why it does not fit alongside fan art and broadcast content under one verb.

3.3 Dimensions that cut across behaviours

Two families of dimension apply across multiple behaviours rather than belonging to any single one: a *behaviour-side* family, with one member so far (institutionalization), and a *person-side* family, with two members kept paired (parasocial attachment and social embeddedness) because, as the paragraphs below argue, they are orthogonal rather than one spectrum.

Institutionalization (behaviour-side) measures how organized and formalized the context of a behaviour-instance is, ranging from none through informal, semi-organized, to fully institutional. It applies natively to Play. It also applies to Watch, Discuss, and Attend — but there it measures the institutionalization of *what is being consumed or attended*, not of the consuming act itself. Watching an individual creator's own broadcast and watching an officially produced tournament broadcast are structurally different experiences even at identical viewing volume, and collapsing that distinction into a single "watch hours" figure — or worse, into a presumed category label for the title being watched — hides exactly the thing that often matters most.

Parasocial attachment and social embeddedness (person-side) are kept in the framework deliberately even though neither is currently measurable from the kind of trace data (viewership logs, transaction records, participation counts) that is typically available. The reasoning for naming an axis that cannot yet be measured is straightforward: a named, unmeasured dimension can be picked up the moment a measurement technique becomes available; a dimension that was never named cannot be retrofitted later without first re-deriving why it matters.

Parasocial attachment is a one-directional relationship between a person and the object of their attachment — who does not know the person exists. Social embeddedness is, by contrast, the degree to which a behaviour is co-experienced with a person's own real, reciprocal relationships. These are not

opposite ends of one spectrum, despite an intuitive pull toward treating them that way. They are **orthogonal**: a person can watch a favourite performer deeply attached and alone, or watch the same performer as a genuinely shared social occasion with their own friend group, or attend a large shared event mainly for the communal occasion with no strong attachment to any specific participant at all. Four real, distinguishable combinations exist, not two.

A further refinement worth recording: a parasocial attachment-object need not be an individual. Devotion to an institution — a team, a brand — is still structurally one-directional, even though the object is an organization rather than a person.

3.4 Centrality and investment

Institutionalization (§3.3) aggregates one behaviour across an entire population to produce a network-level object: a named, persistent institution. There is a structurally identical operation available in the other direction — aggregating *all* of one person's behaviours, across time, to produce an individual-level object. Where institutionalization asks "how organized has this behaviour become, across everyone doing it," this asks "how invested has this person become, across everything they do." The two are transposes of the same underlying behaviour matrix — one read column-wise (per behaviour, across people), the other row-wise (per person, across behaviours) — which is why this construct surfaced directly out of trying to place Socialise (§3.5) rather than being designed in from the start.

We refer to this provisionally as **centrality**, or equivalently **investment**: a person's cumulative position in the network, driven jointly by overall behavioural intensity and by social embeddedness specifically. A person with low-quantity, private, low-embeddedness behaviour across the board sits toward the periphery and likely holds relatively little personal identity in the network; a person with high-quantity, socially embedded behaviour sits nearer the core and likely self-perceives as more central to it.

Like institutions, this state plausibly feeds back into future behaviour — a highly invested member should be more resistant to leaving through a quality dip in the practice, more likely to recruit others, and more likely to defend the community publicly — though this remains a hypothesis carried by the structural parallel to institutionalization's own feedback property (§3.2), not something directly demonstrated.

One further design question is open and worth naming rather than deciding by default: whether centrality/investment should be a purely cumulative quantity (an all-time integral of behaviour) or should **decay** over time absent continued activity. A purely cumulative version cannot express a person who was deeply invested for years and then quietly stopped — it would keep reading them as maximally central indefinitely, which is very likely wrong. A decaying version is also a necessary ingredient for giving the framework any vocabulary for departure from the network at all — see §9's discussion of why the framework currently has none.

3.5 Purchase and Socialise: two behaviours that turned out to be something else

Two further candidates illustrate the value of applying the same collapse test rigorously rather than accepting a familiar-sounding verb at face value.

Purchase looks, at first glance, like an obvious behaviour: people spend money in relation to the network. But examined closely, a purchase is almost always better understood as a *quantified expression* of something else — paid access to another behaviour (a subscription is paid Watch; an entry fee is paid Play), or a monetized signal on the parasocial or social-embeddedness axes (a direct donation to a specific performer is parasocial expression; buying branded apparel to display community membership to one's own social circle is social expression). Spend amount itself is also a less stable unit than it first appears, since purchasing power varies substantially across regions; a comparative framework is better served by counting discrete purchase decisions than by summing currency value across contexts. Purchase survives in the framework as a measurement channel that can be read off several other axes, not as its own behaviour.

Socialise looked, initially, like a natural companion to Discuss — another social verb alongside spectating and talking. It does not survive scrutiny as such, for a structural reason: Discuss is genuinely episodic, a bounded verbal exchange, the same shape as the other four surviving behaviours. Socialise is not. "Becoming socialised within a network" reads more like a state a person enters and holds — private, then embedded — than like a discrete, repeatable act. Pressed further, Socialise turns out to be the person-level mirror of institutionalization itself, applied the other direction across a person's own behaviours rather than across a population's readings on one behaviour. The construct this produces, centrality/investment, is substantial enough to warrant its own treatment — see §3.4.

4. Modes should be discovered, not assumed

It is tempting, once a vocabulary of behaviours exists, to sort networks into a small number of named types — a competitive-practice type, a narrative-consumption type, a social-sandbox type — and then describe each type's characteristic behaviour profile. This framework deliberately resists doing that as a first step. The number and shape of network "types," if they exist at all, should be an **output** of measuring many real networks on the behaviour and institution vectors described above and clustering them, not an **input** chosen in advance and then confirmed by selectively noticing supporting evidence. A cluster found this way should additionally be tested against a null model — compared to what the same clustering procedure would find in randomly rewired data — so that apparent types can be distinguished from patterns that would appear in unstructured data by chance. Named categories are a legitimate and useful endpoint; they are not a legitimate starting assumption.

5. Precedence is an analytical output, not a taxonomic commitment

A separate temptation is to decide, as part of defining the framework, which behaviour is the "prime mover" that drives the others — the intuitive candidate being Play, on the reasoning that one must participate before one can watch, discuss, or attend. This intuition is worth taking seriously but not worth encoding as an axiom. Watching, in particular, has real channels that do not require a pre-existing base of players at all — algorithmic discovery, a single popular broadcaster, spectacle appeal independent of participation — and a practice can plausibly build a substantial audience around high-level performance *before* that performance becomes institutionally organized, not after.

The resolution this paper proposes is to keep the behaviour taxonomy flat and symmetric — no behaviour privileged by definition — and to treat "which behaviour leads" as a **per-network empirical question**, answered by testing lead-lag relationships across behaviour time series once those time series exist. The claim "participation drives everything else" then becomes a testable, falsifiable statement — true for some networks, false or reversed for others — rather than a premise baked into the measurement instrument itself. A framework whose own structure already assumes the answer cannot be used to discover that the answer sometimes differs.

6. What this framework is designed to enable

Four capabilities motivate the design choices above:

Cross-network comparability without presupposed categories. Because the same behaviour vocabulary applies regardless of the underlying practice, two networks that look nothing alike on the surface can be compared directly on measured axes, rather than compared only within an assumed category. This is the framework's single most load-bearing and least-tested claim; §9 closes with the empirical test that would actually validate it.

Emergent typing. Network "types," to the extent they are real, should fall out of clustering real measurements — and be validated against a null model — rather than being assumed and then illustrated.

Precedence and causal analysis. Keeping behaviours as separable, comparably measured quantities makes it possible to ask genuine ordering and causal questions between them — does watching tend to precede or follow institutionalized play, and does the answer vary by practice — questions that collapse into unanswerable the moment behaviours are folded into one another.

Honest accounting of what cannot (yet) be measured. Parasocial attachment and social embeddedness are retained in the framework explicitly, with no proxy forced onto them, so that any analysis built on this framework can state plainly which dimensions it could actually observe and which it could only name. As measurement techniques mature — richer platform data, new

instruments, new consented data sources — previously unmeasurable axes can be picked up without having to first rediscover that they matter.

7. Visualization: a research direction, not yet built

A framework built from a small number of comparable, numeric axes per network invites a direct visual treatment, and it is worth sketching what that could look like, even though building it is a separate undertaking from the conceptual work above.

The basic shape. Each of the core behaviours (Play, Watch, Discuss, Attend, Create) can be drawn as a spoke radiating from a central point representing the network itself — the same underlying idea as a radar or spider chart, extended into three dimensions. Spoke length encodes behavioural intensity; a second visual channel along each spoke — colour, texture, or a layered shell around the spoke's tip — encodes that behaviour's institutionalization reading. The resulting polyhedral shape becomes a network's visual signature: two networks with genuinely different behaviour profiles produce visibly different solids, and because every network is drawn against the *same* fixed set of spokes, shapes are directly comparable by overlaying them, which is precisely the design goal the underlying measurements were built to support.

Representing what cannot be measured. Dimensions this project cannot currently observe — parasocial attachment, social embeddedness, most of Purchase — should not simply be omitted from the picture. Rendered as translucent, dashed, or desaturated spokes at an explicitly marked "unknown" length, they keep the dimension visually present without asserting a value that was never measured. A viewer should be able to tell, at a glance, which parts of a network's shape are real readings and which are acknowledged gaps.

Population distribution, not just network aggregate. Centrality/investment (§3.4) is a per-person quantity, not a per-network one, and could be layered onto the same visualization as a field of points scattered around the solid — points near the core representing highly invested individuals, points toward the periphery representing more occasional participants — turning the shape from a single aggregate summary into a visible distribution of where real people actually sit within it.

Time as a dimension. Because networks evolve, the same shape could animate: a title's polyhedron morphing year by year as institutionalization rises on some spokes and intensity shifts on others, with a simple scrubber allowing a viewer to move through a network's history and watch its signature change shape — a direct, intuitive rendering of the adoption-curve question this framework was built to investigate in the first place.

A longer-term vision. None of the above requires novel rendering technology — standard web-based 3D rendering (WebGL-class tooling) and interactive charting libraries are more than sufficient. The real difficulty is not the rendering layer at all: it is that a five-spoke shape requires five actually-measured behaviours, and §3.1 is explicit that two of the five — Create beyond broadcast activity, and especially

Attend — are barely measurable today. A rendering engine can be built in a sprint; the inputs it would need cannot. The natural long-term endpoint is still an interactive site where any tracked network's shape can be explored, compared side by side with others, and scrubbed through time, with every greyed-out spoke serving as a standing, visible invitation to go measure the thing that spoke represents.

8. Related literature and the paper's actual contribution

This framework was developed independently, from applied work on digital fandom and competitive-practice networks, rather than as a direct application of any single existing theory. Most of its individual components, examined on their own, already have well-developed homes in existing literature, and it is more honest — and more useful to a reader deciding whether to engage with this framework — to say so plainly than to imply novelty where the real contribution lies elsewhere.

The behaviour-over-population-segment starting premise echoes **uses and gratifications** research in media studies (Katz, Blumler & Gurevitch), which asks what functions media consumption serves for an individual rather than treating an audience as an undifferentiated mass. The parasocial attachment dimension traces directly to **Horton and Wohl's** foundational 1956 account of parasocial interaction — a one-directional relationship a person can form with a media performer who does not know they exist — since extended by a growing literature specifically into livestreaming contexts. Social embeddedness, as both a term and a structural idea, is strongly associated with **Granovetter's** account of how a person's ties — strong and weak alike — structure their position and behaviour within a network; using the term without this citation would understate its debt. The institutionalization dimension, and the account of how institutions form out of sustained practice and then reshape the behaviour that produced them, connects to two traditions at once: DiMaggio and Powell's account, in organizational sociology, of how fields professionalize and converge once a critical mass of activity exists to justify it, and — closer to the specific feedback loop this framework depends on, where structure both results from and constrains the ongoing behaviour that sustains it — **Giddens' structuration theory**. The adoption-curve framing in §1 and §5 draws on the classical **diffusion of innovation** literature (Rogers) and the **Bass diffusion model**, which separates adoption into external influence (marketing, media exposure) and internal influence (word of mouth, imitation) — a distinction plausibly relevant to why Watch and Play can decouple, since the two behaviours may be driven by different mixes of these two forces. The competitive-practice behaviour itself — persistent, community-organized, skill-based practice, independent of its delivery substrate — sits close to **communities of practice** as developed by Lave and Wenger, and Create's broader treatment of fan-produced output connects to the **participatory culture** tradition in fandom studies associated with Henry Jenkins. Centrality/investment (§3.4), as a cumulative measure of a person's structural position and stake in a network, sits closer to **Bourdieu's** account of field and capital than to anything else cited here, and is arguably the least-anchored construct in the framework as a result — worth returning to directly rather than treating as adequately covered by the citations above. Finally, the professionalization of competitive gaming specifically has its own dedicated academic treatment,

notably T.L. Taylor's work on the rise of organized esports, which engages directly with the institutionalization question this framework treats as one axis among several rather than the central phenomenon.

Given how much existing ground this covers, the paper's actual claim to contribution should be stated narrowly. It is not a new theory of how communities form, attach, or professionalize — the traditions above already describe those dynamics well. What is different, in descending order of how load-bearing the claim is:

1. **Cross-substrate comparability as a deliberate design goal.** Not the common observation that esports resembles traditional sport, but a vocabulary built specifically so that a Dota scene, a football fandom, and a bridge club yield readings on the *same* axes, comparable directly rather than by analogy. Fandom studies works case by case; sports sociology stays within sport. A genuinely shared instrument, if it holds up, is a real and checkable contribution — and §9 names the empirical test that would check it.
2. **Insisting parasocial attachment and social embeddedness are orthogonal**, with four combinations rather than the two a spectrum framing would imply. This is a crisp, testable claim, and the intuitive collapse into one spectrum is common enough that stating the alternative clearly has independent value.
3. **Naming dimensions the framework cannot currently measure and keeping them visibly present as gaps** (§7's dashed, desaturated spokes at an "unknown" length) rather than either omitting them or forcing a proxy onto them.

In short: this paper's contribution is best understood as **operationalization, not new theory** — taking dynamics the traditions above already describe well and converting several of them into a vocabulary built for measurement from trace data, which most of those traditions were not built for. That claim is testable against the data this project already has; a claim of theoretical novelty would not be.

9. Open questions

This remains a working framework, not a finished one. Items below are ordered by how foundational each is, not by how easy it is to fix.

How a network's boundaries are individuated. The paper describes behaviour and institution readings *of* a network throughout, and never specifies how one network is told apart from another. Is a title's successive competitive-engine versions — an older game and its full successor — one network or several? Is a fragmented set of regional scenes for the same practice one network or many? This is not a minor gap: measuring a thing presupposes being able to say where it starts and stops, and community-detection procedures of the kind this framework's own empirical precursor work relies on are themselves boundary-finding exercises, run without an explicit boundary definition to validate against. This paper does not attempt to resolve it here. One avenue worth naming: the framework's own

tools may bear on the question rather than requiring an answer imported from outside it. If two candidate networks show near-identical behaviour profiles *and* high measured crossover between their participants, that is evidence they are one network rather than two — which would make boundary-detection an output the framework can produce, not a precondition it has to assume before it can be applied at all.

How to model leaving. All five behaviours in §3.1 are acts of doing; institutions have a full lifecycle (§3.2: formation, growth, stabilization, decline); persons, in the current framework, do not. A premise that behaviours are states people "activate into and out of" needs vocabulary for the "out of" half, and currently has none — not lapse, not dormancy, not churn. §3.4 raises one candidate fix, making centrality/investment a decaying rather than purely cumulative quantity, but decay alone may not be enough: a network that is retaining its existing members while failing to recruit new ones, and a network that is actively losing the members it has, are two very different situations that could look identical under a single aggregate decay term.

Whether Attend survives as an independent behaviour or is better modelled as a physical-co-presence context that modifies Play, Watch, or Discuss.

Whether centrality/investment's inputs are complete — whether behavioural intensity and social embeddedness are sufficient, or whether parasocial attachment also contributes independently.

Whether institutionalization is a genuinely independent axis for Purchase or is redundant with the institutionalization of whichever behaviour a given purchase is attached to.

Whether adversarial structure (competitive, cooperative, or solo) and skill level need to be tracked as their own dimensions on Play, separate from institutionalization.

Drafting the behaviour and institution tables for network types beyond competitive-practice — narrative-consumption and social-creative-sandbox candidates in particular — likely a precondition for the emergent-clustering approach in §4 to have more than one mode's worth of data to work with.

Building the per-network, per-behaviour-pair lead-lag analysis described in §5, which requires assembling time series per behaviour that do not yet exist in usable form, and which §3.1's measurability table makes clear will be data-limited for at least two of the five behaviours.

Validating the framework's central claim empirically. Every item above is a refinement within the framework; none of them tests its actual distinguishing promise, which is cross-substrate comparability (§8). Nothing in this paper yet demonstrates that two structurally different networks, measured on the same five axes, produce profiles that differ in an interpretable way — until that is shown, the honest description of this work is a vocabulary, not a validated instrument. The clearest available next step is to measure one esports network and one adjacent non-esports network on identical axes and compare the resulting shapes directly. Chess is the strongest available candidate on the data side: both Lichess and Chess.com publish substantial public play data, a real professional competitive tier exists, and an active streaming scene already exists on Twitch — which would let Play, Watch, and Create all be measured with data this project already knows how to collect. A traditional sport with public

attendance and broadcast-viewership figures is a second worthwhile candidate specifically because it is the one substrate that would exercise Attend, which esports data alone cannot.